Q2) a) i) by using + operator

s1 = "Hello"
s2 = "JAVA"
s3 = s1 + "!" + " " + s2 + "."
print(s3) output: Hello! JAVA.

ii) by join() method

s = ["Hello", "I", "am", "Ankana"
s_new = " ".join(s)
print(s_new)
output: Hello I am Ankana

iii) by using formatted string literal

s1 = "computer"
s2 = "world"
print(f"{s1} {s2}")
output: Computer world

iv) by using format()

s1 = "Ankana"
s2 = "Ghosh"
s_new = "{} {}".format(s1, s2
print(s_new)
o/p = Ankana Ghosh

Paper : DSE-B-2

(Programming Using Python)

Full Marks : 50

Answer ***question no.*** **1** and ***any four*** questions from the rest.

1. Answer ***any five*** questions : 2×5

(a) Define Python interpreter.

(b) Differentiate between mutable datatype and an immutable datatype.

(c) What are the differences between set and dictionary?

(d) What is the use of 'self' keyword in Python?

(e) How does list constructor work?

(f) State the difference between .py and .pyc files.

(g) How are the following statements different from each other?

```
list2 = list1
list2 = list1.copy()
```

list1 is aliased as list2, shallow copy only one list with two diff names.

(h) "An item in a set cannot be accessed through index."— Comment.

2. (a) Discuss the method to join Python strings. Give examples.

(b) Explain the use of split () method in Python.

(c) What is the purpose of `range ()` function? How can you use `range` function to create a list? 4+2+(2+2)

3. (a) Discuss with examples the different modes to open a file in Python.

(b) Write a code snippet to sort a dictionary according to the values.

(c) Explain with an example how exceptions are handled in Python. 3+3+4

4. Consider a Python dictionary of sports which represents information about sports — name, indoor/outdoor type, number of players involved and season of sports.

(a) Create a dictionary with sports name "football", number of players are 11 and type as outdoor.

(b) Add another record for a sports name cricket and other necessary values. 5+5

4) a) dict1 = {'sport1': {'name': 'football',
'no_of_Players': 11,
'type': 'outdoor',
'Season': 'all'}
}

b) dict1['sport2'] = { }
dict1['sport2']['name'] = 'cricket'
dict1['sport2']['no_of_Players'] = 11
dict1['sport2']['type'] = 'outdoor'
}

5. Consider three Python lists as follows :

```
list1 = [1,2,3,4,5]
list2 = [2,4,6,8]
list3 = [1,3,7,3]
res = []
```

(a) What will be the value of `res` after the following code is executed?

```
res.extend (i for i in list1 if i not in (list2 + list3))
```

5

(b) Write a code snippet to display all elements in `list2` which are not in `list1`.

(c) Write a code snippet to display the duplicate elements of `list3`.

(d) Write a Python function that accepts a list as an argument and returns true if the list is an empty list and false otherwise. 2+2+2+4

6. (a) Write Python statements to print the following regarding current date and time :

- Date and time
- Year
- Day of the week (e.g. Sunday)
- Month name

super() is a built in method which is useful to call the superclass constructor or method from the subclass.
See P-24 from Notes

(b) Write the purposes of `super` keywords.

(c) Write Python statements to perform the following tuple operations :

- Print all items from the third position to the end.
- Add a tuple to the end of another tuple.
- Loop through a tuple.
- Multiply contents of a tuple. 4+2+4

7. (a) What are the differences between 'pass' and 'continue' statement in Python?

(b) "try block can't exist without any except block." — Justify your answer.

(c) str = 'Python is very easy programming!!!'. What is the output of print(str[:7]) and print(str[–5]) and print(str[: :])? 3+4+3

8. (a) What is range() function and how it is used in lists? Give the code snippet.

(b) Write a program that copies one Python script into another in such a way that all comment lines are skipped and not copied in the destination file.

(c) How does list can be used as QUEUE data structure with a proper code snippet? 3+4+3

[P-332, R.Thereja]

Q5)b) res = [i for i in list2 if i not in list1]
print(res)

c) duplicates = [i for i in list3 if list3.count(i) > 1]
print("Duplicate elements", duplicates)

<PTO>

5) d) def is_empty(list1):

return(len(list1) == 0) # returns

True if the list1 is empty, False otherwise

print(is_empty([])) # output True

print(is_empty([1, 2, 3])) # output False

Q6) a) from datetime import * (or import datetime)

get current date & time

now = datetime.now()

print("Current Date & Time:", now)

print("Year:", now.year)

print("Day of the week:",

now.strftime("%A"))

eg: Sunday

print("Month Name:", now.strftime("%B"))

eg: February

Q6) c) tup = (10, 20, 30, 40, 50, 60)

print(tup[2:]) # output: (30, 40, 50, 60)

tuple1 = (1, 2, 3)

tuple2 = (4, 5, 6)

Adding tuple2 to the end of tuple1

result = tuple1 + tuple2

print("Resultant Tuple:", result)

my_tuple = (10, 20, 30, 40, 50, 60)

loop through the tuple

for item in my_tuple:

    print(item)

my_tuple = (1, 2, 3)

result = tuple(i * 3 for i in my_tuple)

print(result)

Q3/5) Dictionary Sort

d = {2:56, 1:2, 5:12, 4:24}

print("Dictionary:", d)

sorting by keys

for i in sorted(d.keys()):

    print(i, end=" ")

sorting by values

```
for i in sorted(d.values()):
    print(i, end=" ")
```

OR

by using lambda fn

```
my_dict = {"apple": 5, "banana": 10, "cherry": 3}
sorted_dict = dict(sorted(my_dict.items(),
                          key = lambda item:
                                item[1]))
print(sorted_dict)
```